

Topic 6: The Carbon Cycle and Energy Security

Overview

A balanced carbon cycle is important in maintaining planetary health. The carbon cycle operates at a range of spatial scales and timescales, from seconds to millions of years. Physical processes control the movement of carbon between stores on land, the oceans and the atmosphere. Changes to the most important stores of carbon and carbon fluxes are a result of physical and human processes. Reliance on fossil fuels has caused significant changes to carbon stores and contributed to climate change resulting from anthropogenic carbon emissions.

The water and carbon cycles and the role of feedbacks in and between the two cycles, provide a context for developing an understanding of climate change.

Anthropogenic climate change poses a serious threat to the health of the planet. There is a range of adaptation and mitigation strategies that could be used, but for them to be successful they require global agreements as well as national actions.

Content

Enquiry question 1: How does the carbon cycle operate to maintain planetary health?

Key idea	Detailed content
6.1 Most global carbon is locked in terrestrial stores as part of the long-term geological cycle (slow carbon cycle).	a. The biogeochemical carbon cycle consists of carbon stores of different sizes (terrestrial, oceans and atmosphere), with annual fluxes between stores of varying size (measured in Pg/Gt), rates and on different timescales. (1)
	b. Most of the earth's carbon is geological, resulting from the formation of sedimentary carbonate rocks (limestone) in the oceans and biologically derived carbon in shale, coal and other rocks.
	c. Rainwater becomes a weak carbonic acid when it absorbs carbon dioxide from the atmosphere that then reacts with silicate minerals. This chemical weathering process releases ions such as calcium, which are transported by rivers to the oceans, where organisms combine calcium with dissolved carbon to create calcium carbonate. Carbonate rock forms via sedimentation of dead organisms, and carbon dioxide is released back into the atmosphere by volcanism.